Extra Dessert #7: Space and Earth Stations

Telemetry, telecommand, balloon ID, band allocations for space stations, model-craft power limits, and one-way ops.

Group E1D · 12 pool questions · Self-study reference · Practice exams at eham.net/exams

Definitions

Term	Definition	Q
Telemetry	**One-way transmission of measurements at a distance from the measuring instrument**	E1D01
Space telecommand station	Amateur station that transmits communications to **initiate, modify, or terminate functions of a space station**	E1D03

Encryption

Question	Answer	Q
Who may transmit encrypted messages?	**Space telecommand stations** (encryption allowed to prevent unauthorized control of a satellite)	E1D02

Identification And Posting

Question	Answer	Q
Required ID in a balloon-borne telemetry station's transmissions	**Call sign**	E1D04
What must be posted at a telecommand station within 50 km of Earth's surface	Photocopy of station license + label with licensee name/address/phone + label with control operator name/address/phone – **all of the above**	E1D05

Power And Band Allocations

Topic	Answer	Q
Max output power for model-craft telecommand	**1 watt**	E1D06
HF bands authorized for space stations	**40 m, 20 m, 15 m, and 10 m**	E1D07
VHF band(s) authorized for space stations	**2 meters only**	E1D08
UHF bands authorized for space stations	**70 cm and 13 cm**	E1D09

Eligibility

Question	Answer	Q
Who may be a telecommand station of a space station?	**Any amateur station so designated by the space station licensee**, subject to control op privileges	E1D10
Who may operate as an Earth station?	**Any amateur station, subject to the privileges of the class of operator license held by the control operator**	E1D11

One-Way Communications

Question	Answer	Q
Which stations may transmit one-way?	**A space station, beacon station, or telecommand station**	E1D12

Dessert 7 Question Bank (E1D)

E1D01. What is the definition of telemetry?
A. One-way transmission of measurements at a distance from the measuring instrument.
B. Two-way transmissions in excess of 1000 feet.
C. Two-way transmissions of data.
D. One-way transmission that initiates, modifies, or terminates the functions of a device at a distance.

E1D02. Which of the following may transmit encrypted messages?
A. Telecommand signals to terrestrial repeaters.
B. Telecommand signals from a space telecommand station.
C. Auxiliary relay links carrying repeater audio.
D. Mesh network backbone nodes.

E1D03. What is a space telecommand station?
A. An amateur station located on the surface of the Earth for communication with other Earth stations by means of Earth satellites.
B. An amateur station that transmits communications to initiate, modify, or terminate functions of a space station.
C. An amateur station located in a satellite or a balloon more than 50 kilometers above the surface of the Earth.
D. An amateur station that receives telemetry from a satellite or balloon more than 50 kilometers above the surface of the Earth.

E1D04. Which of the following is required in the identification transmissions from a balloon-borne telemetry station?
A. Call sign.
B. The output power of the balloon transmitter.
C. The station's six-character Maidenhead grid locator.
D. All these choices are correct.

E1D05. What must be posted at the location of a station being operated by telecommand on or within 50 kilometers of the Earth's surface?
A. A photocopy of the station license.
B. A label with the name, address, and telephone number of the station licensee.
C. A label with the name, address, and telephone number of the control operator.
D. All these choices are correct.

E1D06. What is the maximum permitted transmitter output power when operating a model craft by telecommand?
A. 1 watt.
B. 2 watts.
C. 5 watts.
D. 100 watts.

E1D07. Which of the following HF amateur bands include allocations for space stations?
A. 40 meters, 20 meters, 15 meters, and 10 meters.
B. 30 meters, 17 meters, and 10 meters.
C. Only 10 meters.
D. Satellite operation is permitted on all HF bands.

E1D08. Which VHF amateur bands have frequencies authorized for space stations?
A. 6 meters and 2 meters.
B. 6 meters, 2 meters, and 1.25 meters.
C. 2 meters and 1.25 meters.
D. 2 meters.

E1D09. Which UHF amateur bands have frequencies authorized for space stations?
A. 70 centimeters only.
B. 70 centimeters and 13 centimeters.
C. 70 centimeters and 33 centimeters.
D. 33 centimeters and 13 centimeters.

E1D10. Which amateur stations are eligible to be telecommand stations of space stations, subject to the privileges of the class of operator license held by the control operator of the station?
A. Any amateur station approved by AMSAT.
B. Any amateur station so designated by the space station licensee.
C. Any amateur station so designated by the ITU.
D. All these choices are correct.

E1D11. Which amateur stations are eligible to operate as Earth stations?
A. Any amateur licensee who has successfully completed the AMSAT space communications course.
B. Only those of General, Advanced or Amateur Extra class operators.
C. Only those of Amateur Extra class operators.
D. Any amateur station, subject to the privileges of the class of operator license held by the control operator.

E1D12. Which of the following amateur stations may transmit one-way communications?
A. A space station, beacon station, or telecommand station.
B. A local repeater or linked repeater station.
C. A message forwarding station or automatically controlled digital station.
D. All these choices are correct.

Answer Key

E1D01: A	E1D02: B	E1D03: B	E1D04: A
E1D05: D	E1D06: A	E1D07: A	E1D08: D
E1D09: B	E1D10: B	E1D11: D	E1D12: A